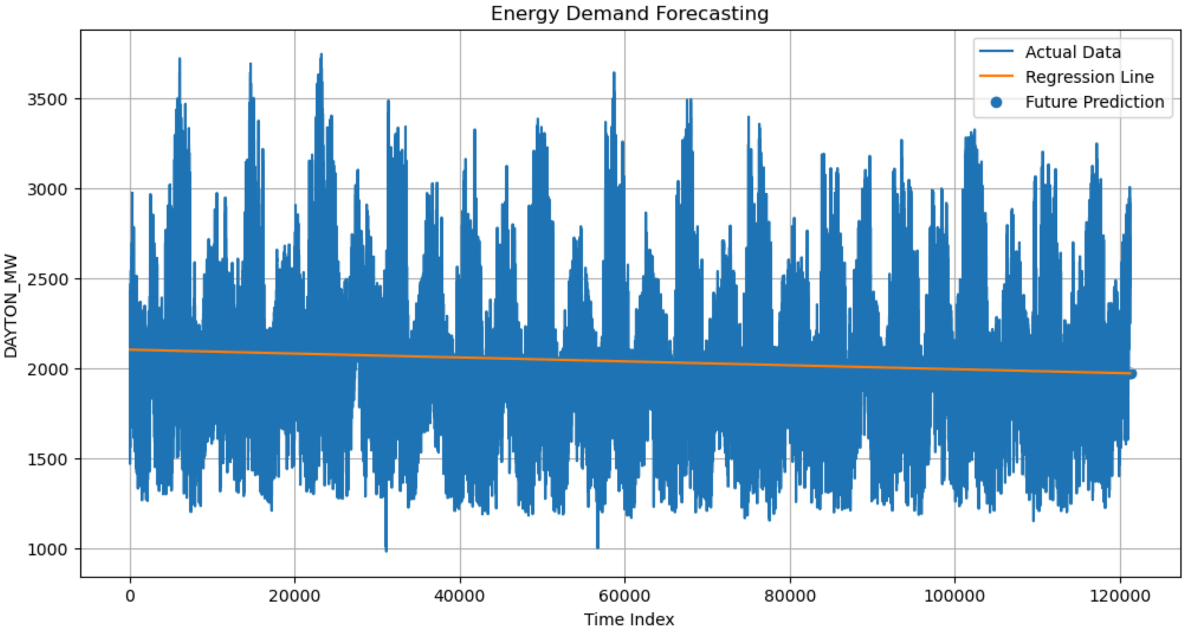
In [2]:
```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
path = r"C:\Users\Dhanashri\Downloads\Energy Demand Forecasting\DAYTON_hourly.csv"
try:
    data = pd.read_csv(path)
    print("Columns in dataset:")
    print(data.columns)
    numeric_cols = data.select_dtypes(include='number').columns
    if len(numeric_cols) == 0:
        print("No numeric column found in dataset")
    else:
        target_col = numeric_cols[0]
        print("Using column:", target_col)
        data['TimeIndex'] = range(len(data))
        X = data[['TimeIndex']]
        y = data[target_col]
        model = LinearRegression()
        model.fit(X, y)
        future_steps = pd.DataFrame({
            'TimeIndex': [len(data), len(data) + 1]
        })

        pred = model.predict(future_steps)
        print("Predicted values:")
        print(pred)
        plt.figure(figsize=(12,6))

        plt.plot(data['TimeIndex'], y, label='Actual Data')
        plt.plot(
            data['TimeIndex'],
            model.predict(X),
            label='Regression Line'
        )

        plt.scatter(
            future_steps['TimeIndex'],
            pred,
            label='Future Prediction'
        )
        plt.xlabel("Time Index")
        plt.ylabel(target_col)
        plt.title("Energy Demand Forecasting")
        plt.legend()
        plt.grid(True)
        plt.show()
except FileNotFoundError:
    print("File not found. Check the path.")
except Exception as e:
    print("Error:", e)
```

```
Columns in dataset:
Index(['Datetime', 'DAYTON_MW'], dtype='object')
Using column: DAYTON_MW
Predicted values:
[1972.07530711 1972.07422238]
```


Energy Demand Forecasting
Actual Data
Regression Line
Future Prediction
DAYTON_MW
1000
1500
2000
2500
3000
3500
0
20000
40000
60000
80000
100000
120000
Time Index

```
In [ ]:
```